

YASHAS CHANDRA BATHINI AI Engineer | Full-Stack Developer | LLM & RAG Systems | Azure & CI/CD

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 yashas144.github.io/web  F-1 OPT Eligible (June 2026)

PROFILE

AI Engineer with **2+ years of experience** building scalable LLM-powered applications and RAG systems, backed by strong full-stack expertise in React and FastAPI. Delivered end-to-end AI platforms including intelligent predictive maintenance systems, recommendation engines and real-time analytics systems. Deep expertise in hybrid retrieval (BM25 + vector search), NLP pipelines, and explainable AI. Proven ability to design high-performance APIs, deploy cloud-native systems, and ship responsive, user-centric products. Experienced across the full AI lifecycle—from data pipelines and model integration to production deployment and monitoring.

SKILLS

Generative AI & LLMs: Transformers, Hugging Face, RAG, Agentic AI, LangChain, NLP, FAISS, Vector DB (Chroma/Pinecone).

Full-Stack Web Development: React (Frontend), Python (FastAPI), REST APIs, LLM Integration, scalable API development.

Programming: Python, Java, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn.

Data Engineering: Data Analysis, Data Visualization, Tableau, MySQL, Snowflake.

MLOps: Regression, Clustering, Time-Series Forecasting (LSTM, AR models), Anomaly Detection (Isolation Forest), XGBoost / LightGBM

Cloud & DevOps: Azure App Services, AWS, CI/CD Pipelines (GitHub Actions), Docker, Git, Jira, VS Code, Jupyter, Google Colab

PROJECT EXPERIENCE

Auravance - The Future of Industrial Foresight 06/2025 – 06/2026

- Auravance is an AI-powered predictive maintenance and industrial automation platform that senses equipment failures before they happen — keeping factories running smarter, longer, and without interruption.
- Building a full-stack AI platform (React, FastAPI) for real-time equipment monitoring and failure prediction using XGBoost and LSTM, improving detection accuracy by 30%.
- Developing a RAG-based root cause analysis system over equipment manuals, generating explainable diagnostics.
- Designing RUL (Remaining Useful Life) prediction and anomaly detection (Isolation Forest) to reduce downtime risk.
- Engineering scalable real-time IoT data pipelines using **MQTT** and Python to ingest, process, and store high-frequency sensor telemetry, enabling anomaly detection, predictive forecasting, and operational insights.
- Implementing proactive automated maintenance scheduling, alerting workflows, and operational dashboards.

AI-Powered Music Intelligence Platform. Live URL:  07/2024 – 05/2025

- Designed and delivered an AI-powered music recommendation platform leveraging Hybrid-RAG (BM25 + vector search) across 500K+ songs, enabling context-aware discovery based on mood, genre, and user intent.
- Architected a scalable RAG pipeline with embedding-based retrieval and ranking, significantly improving recommendation relevance and reducing noise in outputs.
- Integrated LLM-driven explainability, enhancing user trust and transparency in AI recommendations.
- Engineered real-time features, including real-time mood detection and seamless playback.
- Deployed a cloud-native full-stack system (React + FastAPI) on Azure and Hugging Face with CI/CD pipelines, ensuring reliable and scalable delivery.

WORK EXPERIENCE

AI Engineer 06/2022 – 07/2024
Sysarket
Hyderabad, India

Deep Learning-Based Image Synthesis with GAN Architecture. Research Paper URL: 

- Developed GAN-based image synthesis systems using PyTorch and TensorFlow, improving output quality through architecture tuning and training optimization.
- Contributed to applied generative AI research, developing GAN architectures for real-world applications and publication-ready experimentation.
- Collaborated in agile teams to deliver scalable ML solutions and optimize end-to-end training pipelines.

AI-Based Sports Analytics Platform

- Developed a robust sports analytics platform featuring ML-based match predictions, APIs, and dashboards.
- Improved model accuracy by 18% using feature engineering, hyperparameter tuning, and MLflow tracking.
- Implemented model registry, versioning, and monitoring (performance, confidence, and feature drift)
- Deployed a containerized full-stack application using React (frontend) and Flask (backend) with integrated ML services, leveraging Docker and Azure with Azure SQL DB for scalable, production-ready model serving.

EDUCATION

Master of Science - Data Science 08/2024 – 05/2026
University of North Texas, GPA: 4.0 / 4.0
Denton, Texas

Bachelor of Technology - Information Technology 08/2020 – 05/2024
Jawaharlal Nehru Technological University, Hyderabad, CGPA: 3.7 / 4
Hyderabad, India

CERTIFICATIONS

Generative AI Fundamentals — Databricks | **Machine Learning with Big Data** — UC San Diego | **Introduction to Data Science** — Cisco | **Career Essentials in Generative AI** — Microsoft | **Python Essentials - 1** — Cisco